

Nonexistence of integer solutions to $6 + x^3 - y^2 + y^2z^2 = 0$

Theorem

There are no integers x, y, z with z even, y odd, and $x \equiv 3 \pmod{4}$ satisfying

$$6 + x^3 - y^2 + y^2z^2 = 0.$$

Proof

Rewrite the equation as

$$x^3 + 6 = -y^2(z^2 - 1). \quad (1)$$

Set $X := -x$. Since the right-hand side of (1) is negative when $y \neq 0$ and $|z| \geq 2$, any solution with $y \neq 0$ must satisfy $x^3 + 6 < 0$. Thus $x \leq -3$, and because $x \equiv 3 \pmod{4}$ we have $x \leq -5$ and hence $X \geq 5$ with $X \equiv 1 \pmod{4}$.

Rewrite (1) as

$$X^3 + 6 = y^2(z^2 - 1), \quad (2)$$

where $X \in \mathbb{Z}_{>0}$, $X \equiv 1 \pmod{4}$, y is odd, and z is even.

1) Coprimality of X and y . Suppose a prime p divides both X and y . Since y is odd, p is odd. Then p divides $X^3 + 6$ and therefore divides 6. Thus $p = 3$. Write $X = 3X'$ and $y = 3y'$. Substituting in (2) gives

$$27X'^3 + 6 = 9y'^2(z^2 - 1).$$

Dividing by 3 yields

$$9X'^3 + 2 = 3y'^2(z^2 - 1).$$

The left-hand side is congruent to 2 modulo 3, while the right-hand side is congruent to 0 modulo 3. This is impossible. Hence no prime divides both X and y , i.e.

$$\gcd(X, y) = 1. \quad (3)$$

2) Parity and modulo 4 properties. Because z is even, both $z - 1$ and $z + 1$ are odd. Their product $z^2 - 1$ is therefore odd, and $y^2(z^2 - 1)$ is odd. In particular, $X^3 + 6$ is odd, so X is odd. Combined with $X \equiv 1 \pmod{4}$, we obtain

$$X + 2 \equiv 3 \pmod{4}, \quad X^2 - 2X + 4 \equiv 3 \pmod{4}. \quad (4)$$

The factorization

$$X^3 + 6 = (X + 2)(X^2 - 2X + 4) \quad (5)$$

takes the form of two coprime odd factors. Indeed,

$$\begin{aligned}\gcd(X+2, X^2-2X+4) &= \gcd(X+2, (X^2-2X+4) - (X-4)(X+2)) \\ &= \gcd(X+2, 12).\end{aligned}$$

Because $X \equiv 1 \pmod{4}$, the number $X+2$ is an odd integer congruent to 3 modulo 4. Thus $\gcd(X+2, 12)$ divides 3.

If $3 \mid X+2$, then $X \equiv 1 \pmod{3}$ and hence $X^3+6 \equiv 1 \pmod{3}$. But z is even, so $z \equiv 0$ or $2 \pmod{3}$, and therefore $z^2-1 \equiv 2$ or $0 \pmod{3}$. Since y is odd, $y^2 \equiv 1 \pmod{3}$. Thus the right-hand side of (2) is congruent to 2 or 0 modulo 3, contradicting $X^3+6 \equiv 1 \pmod{3}$. Consequently $3 \nmid X+2$ and we have

$$\gcd(X+2, X^2-2X+4) = 1. \quad (6)$$

3) The parity of $z \pm 1$. Since z is even, $z-1$ and $z+1$ are odd and coprime. One of them is congruent to 1 modulo 4 and the other to 3 modulo 4. (Indeed, any even z satisfies $z \equiv 0$ or $2 \pmod{4}$, so $z-1 \equiv 3$ and $z+1 \equiv 1$, or vice versa.)

Because y is odd, $y^2 \equiv 1 \pmod{4}$. Therefore any odd integer obtained by multiplying one of $z-1$ or $z+1$ by y^2 has the same residue class modulo 4 as the original factor. In particular,

$$\{(z-1)y^2, (z+1)y^2\} \equiv \{1, 3\} \pmod{4}. \quad (7)$$

4) Final contradiction. Equation (2) can be rewritten using (5) as

$$(X+2)(X^2-2X+4) = y^2(z-1)(z+1). \quad (8)$$

The left-hand side is a product of two coprime odd integers by (6). Since $z-1$ and $z+1$ are coprime, the prime divisors of $(z-1)(z+1)$ must split between the two coprime left factors without overlap. Because y^2 is a square and every odd square is congruent to 1 modulo 4, the residue of each factor modulo 4 is determined entirely by the odd part coming from either $z-1$ or $z+1$.

One of $z-1$ and $z+1$ is $1 \pmod{4}$ and the other is $3 \pmod{4}$. Hence in any factorisation of $y^2(z-1)(z+1)$ into two coprime odd factors, one of the resulting factors must be congruent to 1 modulo 4 and the other to 3 modulo 4. This follows from the fact that an odd square is 1 modulo 4, so the parity class of each factor is unchanged by the square multiplier.

However, from (4) both factors on the left-hand side of (8) are congruent to 3 modulo 4. This is impossible if the right-hand side splits as one factor congruent to 1 and one factor congruent to 3 modulo 4. Therefore no integer solution can exist.

5) Conclusion. The assumptions z even, y odd, and $x \equiv 3 \pmod{4}$ force $x \leq -5$, $X = -x \geq 5$, $X \equiv 1 \pmod{4}$, and coprimality $\gcd(X, y) = 1$. The factorisation argument above then yields a rigid mod-4 contradiction. Hence the original equation has no integer solution under the given hypotheses.